

Discussion Assignment 4

Machine Learning

MATH 2210, Spring 2018

Background

In practice, solutions to many theoretical and real-world problems often rely quite heavily on the mathematical notion of the gradient. For example, situations frequently arise – for instance, constructing a road through the mountains, or planning the flow of water across a landscape – where we are interested in knowing the direction in which a function is increasing or decreasing most rapidly. We have seen such situations occur in connection to machine learning as well. For example, in the video we watched in class, the programmer of a learning computer is interested in minimizing a cost function used in training a particular neural network. Truly minimizing this cost function generally involves what is known as “solving a normal equation,” which is possible, but rarely done in practice (mainly because it is so computationally expensive). Instead, programmers often prefer to use a minimization method known as “gradient descent.” Below is a chart that explores some of the benefits of using one method versus the other:

Gradient Descent	Normal Equation
Needs many iterations	No Iterations
n^2 calculations for n unknowns	n^3 calculations for n unknowns
Works well when n is large	Slow when n is large

The neural network we saw in the video had 13,002 parameters, but for our illustrative purposes, we will work with a cost function associated with a neural network that has only two parameters.

Question

Consider the cost function

$$c(w_1, w_2) = w_1^2 - 10w_1 + 4w_2^2 - 20w_2 + 51.$$

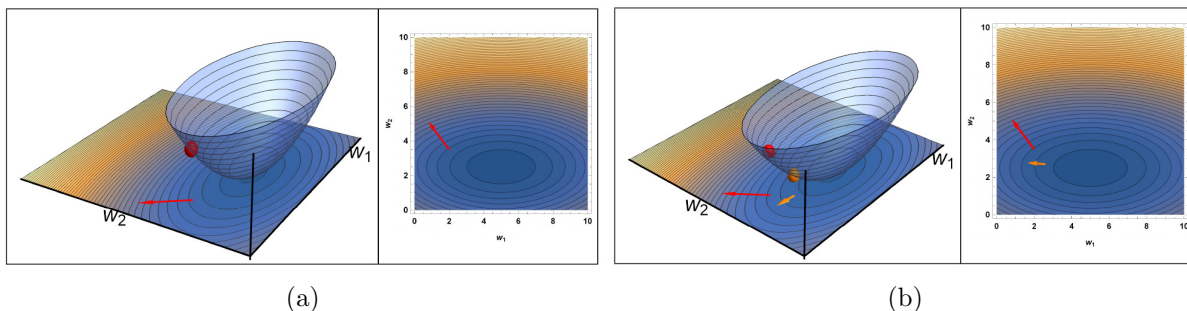


Figure 1: Visualization of the points $(2, 3.5, 14)$ and $(2.6, 2.7, 6.92)$ on the surface of $c(x, y)$.

Lets say you initially set the weights as $w_1 = 2$ and $w_2 = 3.5$. Then, $c(2, 3.5) = 14$. You may visualize this as the point $(2, 3.5, 14)$ which sits on the surface of the graph of c , as is depicted in Figure 1a. In an attempt to minimize the cost function, you can step in the opposite direction of the gradient (that is, you can step in the opposite direction of the rate of change at that point). To ensure that your step size is not too large (that is, to prevent you from overshooting, in a sense), you could choose to step only one tenth the length of the gradient. If you do so, you would choose the weights $w_1 = 2.6$ and $w_2 = 2.7$ and get $c(2.6, 2.7) = 6.92$. You may visualize this as the point $(2.6, 2.7, 6.92)$ which sits on the surface of the graph of c as in Figure 1b. If this process continues, what are the weight values after 10 steps? Choose the best answer from the selection below.

- (a) $w_1 = 4.3, w_2 = 2.7$
- (b) $w_1 = 4.5, w_2 = 2.5$
- (c) $w_1 = 4.6, w_2 = 2.5$
- (d) $w_1 = 4.9, w_2 = 2.5$

Group Work

The group work portion of your grade is worth 3 points. As a group you are allowed to submit three answers. However, incorrect answers will be counted as wrong. For example, if the correct answer is (d) and your group submits $(a), (d), (d)$ as their answers the score will be $2/3$ on this portion of the assignment.

Individual Work

Now you are to write a report, one page maximum, that convinces an audience of your peers that you understand the gradient method and that you know what the weights would be after 10 iterations. This report will need to be saved as a pdf and submitted to your discussion sections WyoCourses page. A scoring rubric and guideline on writing mathematics properly is posted on the WyoCourses page.